

MALAK ESSAM

Backend Developer

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malak-essam Malak-Essam

PROFESSIONAL SUMMARY

Java Backend Developer with hands-on experience building and deploying production-ready applications using Spring Boot. Developed real-time chat application with WebSocket, implemented JWT authentication, and designed PostgreSQL database schemas. Passionate about clean code, software engineering principles, and continuous learning. Seeking an opportunity to contribute to a development team while expanding technical expertise in backend systems.

EXPERIENCE

Software Engineer

Arkan Tech

Oct 2025 – Present Remote, Jordan

- Engineered and maintained reliable RESTful APIs and core backend services utilizing Java and the Spring Boot framework
- Designed and implemented microservices architectures to ensure system scalability, maintainability, and seamless communication between application components
- Optimized database operations and refined business logic to enhance overall application performance and minimize response times
- Collaborated on developing secure enterprise solutions, focusing on stable architecture, secure data transmission, and mitigating operational risks

Programming Instructor

Freelance

July 2025 – Present Remote

- Designed and delivered practical programming projects covering OOP fundamentals, database concepts, and backend development principles
- Mentored students through hands-on coding exercises and real-world application development

Machine Learning Instructor

Cairo University – HCIA Internship

July 2025 – Sept 2025 Hybrid

- Conducted applied machine learning sessions focused on data-driven problem solving and model development using Python
- Guided students through practical ML projects and algorithm implementation

EDUCATION

B.Sc. in Computer Science

Cairo University

Sept 2018 – June 2022 Cairo, Egypt

TECHNICAL SKILLS

Backend Development

Java 21 Spring Boot Spring Security

Spring Data JPA Spring MVC

Hibernate RESTful APIs WebSocket

JWT RabbitMQ

Databases

PostgreSQL MySQL SQL

Database Design Spring Data JPA

Testing & Quality

JUnit Mockito Integration Testing

Unit Testing

Frontend

React Next.js JavaScript

TypeScript HTML/CSS Tailwind CSS

Tools & DevOps

Git Docker Maven Postman

Railway Vercel Linux

Software Engineering

OOP Design Patterns

SOLID Principles Clean Code

API Design Microservices

Full Stack Developer

Egyptian Army

June 2023 – June 2024 Egypt

- Maintained and enhanced web applications using HTML, CSS, JavaScript, and PHP
- Fixed bugs, optimized frontend responsiveness, and implemented security improvements
- Applied IT skills for troubleshooting, deployment, and version control management

PROJECTS

Real-Time Chat Application with Friend Management

Personal Full-Stack Project

Backend: Spring Boot, Spring Security, Spring Data JPA, WebSocket (STOMP/SockJS), PostgreSQL, JWT, RabbitMQ

Frontend: Next.js 16, Tailwind CSS

Backend | Frontend

- Developed full-stack real-time chat application from scratch, implementing user authentication, messaging, and friend management features
- Built RESTful API with 15+ endpoints using Spring Boot for user management, authentication, and chat operations
- Implemented real-time bidirectional messaging using WebSocket (STOMP/SockJS) with typing indicators and online presence tracking
- Integrated RabbitMQ message broker for asynchronous notification delivery and offline message queuing
- Designed PostgreSQL database schema with proper relationships and constraints for users, messages, and friendships
- Secured application with Spring Security and JWT-based authentication including token refresh mechanism and BCrypt password encryption
- Deployed backend to Railway with PostgreSQL on Neon, and frontend to Vercel with automated CI/CD via GitHub integration
- Created comprehensive test suite using JUnit and Mockito to ensure reliability

CERTIFICATIONS



Linux Essentials
MaharaTech – 2025



Database Fundamentals
MaharaTech – 2024

LANGUAGES

Arabic



English



ADDITIONAL INFORMATION

Completed military service in the Egyptian Armed Forces (2024) with appreciation as a good role model.